Supplemental Material

S.I. Legal Document Collection

Detailed information about the legal documents collected for the DriveLegal dataset construction across multiple jurisdictions is provided in Table S1.

Table S1: Legal documents collected for DriveLegal dataset construction across multiple jurisdictions.

Region / Jurisdiction	Language	Official Legal Document Title
Ontario, Canada	English	*Highway Traffic Act* (R.S.O. 1990, c. H.8) ontario.ca/laws/statute/90h08
Michigan, USA	English	*Michigan Vehicle Code* (Act 300 of 1949) legislature.mi.gov (Act 300 of 1949)
Hong Kong SAR, China	English / Chinese	*Road Users' Code* td.gov.hk/road_users_code
United Kingdom	English	*The Highway Code* highwaycodeuk.co.uk
People's Republic of China	Chinese (Simplified)	中华人民共和国道路交通安全法实施条例 *Regulations on the Implementation of the Road Traffic Safety Law* gov.cn/gongbao (2019)
Republic of Korea	Korean	도로교통법 *Road Traffic Act* law.go.kr (Road Traffic Act)
Japan	Japanese	道路交通法 *Road Traffic Act of Japan* japaneselawtranslation.go.jp

S.II. Dataset Details

S.II.A. DriveLegal-SFT

We split the dataset into training and testing sets. Table S2 summarizes the number of samples and the average word counts for each split.

To examine dataset coverage, each instruction was decomposed into verb–noun pairs, and their frequency distribution was visualized using a radial chart. The analysis revealed a balanced mix of instruction types such as explain and discuss and a broad range of legal concepts including penalty, regulation, and rule. This distribution indicates that DriveLegal-SFT encompasses both procedural and conceptual dimensions of legal reasoning, supporting model generalization across jurisdictions and task categories. Sample dataset entries are presented in Table S3.

S.II.B. DriveLegal-RAG

This dataset consists of vector-retrieval-oriented QA pairs derived from driving regulations across multiple jurisdictions and graph-based RAG entries constructed from triplets extracted from legal texts. Sample dataset entries from vector-retrieval and graph-based RAG splits are presented in Table S4 and S5.

Simple RAG. This split serves as the foundational multilingual benchmark for evaluating RAG performance across jurisdictions. In total, **942** entries are constructed, comprising **642 English** entries and **300 multilingual** ones. Among the English entries, **84** originate from Hong Kong, **243** from Michigan, **252** from Ontario, and **63** from the United Kingdom. Additionally, there are **100 entries each** in **Chinese**, **Korean**, and **Japanese**.

Table S2: Statistics of the DriveLegal-SFT dataset. Word counts represent the average number of words per instruction and response.

Split	Samples	Avg. Instruction	Avg. Response
Train	4,318	19.8	86.4
Test	1,080	20.5	88.1
All	5,398	19.9	86.8

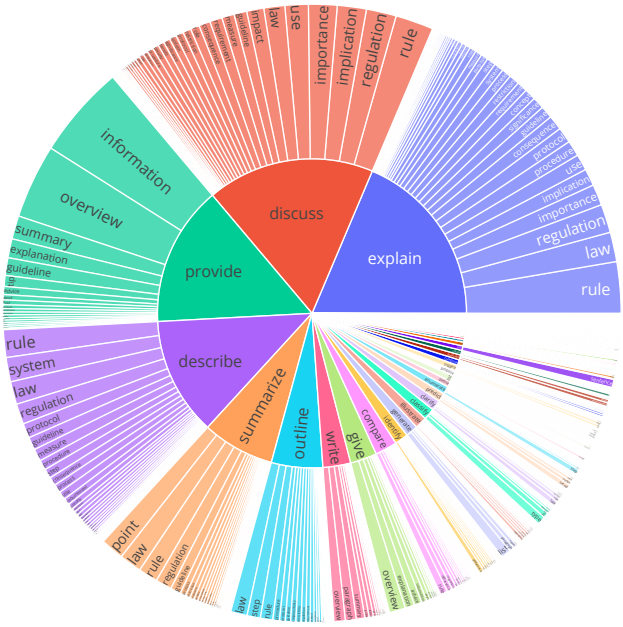

Figure S1: Distribution of the DriveLegal-SFT dataset.

GraphRAG. This split targets higher-complexity retrieval and reasoning on the knowledge graph. It currently contains **500** entries per region (Hong Kong, Michigan, Ontario, United Kingdom), totaling **2000** entries. Tasks are distributed across **Fact Retrieval (42%)**, **Complex Reasoning (30%)**, **Contextual Summarization (20%)**, and **Creative Generation (8%)**.

S.III. Implementation Details

S.III.A. Hardware

Experiments presented in this paper are run on a single H200 GPU. Real-time performance and all case studies are conducted on a desktop with an Intel 13700K CPU, 128 GB RAM and dual RTX 4090 GPUs.

S.III.B. Fine-tuning Configuration

All fine-tuning is run on a dual RTX 4090 machine using the LLaMA-Factory [2] framework. For the Qwen3 model, even with the think flag turned off, the model may still return think block. For fair comparison, all think blocks are trimmed before metrics calculation.

S.III.C. RAG Configuration and Sample Knowledge Graph

For benchmarking, we implement local/global/hybrid mode via LightRAG with parameters shown in Table S8. For deployment, we use a custom implementation to have full control over the hybrid RAG process. Vector and knowledge graph retrievals are deployed as two separate services using FastAPI. Sample knowledge graphs for Ontario, Canada, China, and Japan are shown in Figures S2, S3, and S4.

Instruction	Reference response (truncated)
Provide guidelines on what to do when encountering roadworks or construction zones in ontario, canada.	When encountering roadworks or construction zones in Ontario, Canada, follow these guidelines: 1. **Obey Signs and Signals**: Pay attention to all posted signs, signals, and flag persons directing traffic. 2. **Reduce Speed**: Slow down as you approach the ...
Discuss the importance and rules of yielding to emergency vehicles in south korea.	In South Korea, yielding to emergency vehicles is crucial for ensuring the safety and efficiency of emergency response services. When an emergency vehicle approaches with flashing lights or sirens, drivers must immediately pull over to the right side of the road and stop, allowing the vehicle to pass quickly and safely. ...
Discuss the rules regarding right-of-way at roundabouts in australia.	In Australia, the rules for right-of-way at roundabouts are as follows: 1. **Give Way to Vehicles Already in the Roundabout**: When approaching a roundabout, you must give way to all vehicles already on the roundabout. 2. **Use Indicators Appropriately**: ...

Table S3: Sample entries from the DriveLegal-SFT instruction-tuning dataset.

S.III.D. Evaluation Configuration

For benchmarking, we truncate the context length down to 20,000/30,000 characters. For evaluation, GPT4o-mini is used with a fixed temperature of 0.0.

S.IV. Additional RAG results

S.IV.A. Sensitivity Analysis

As mentioned in the automated evaluation section, we further performed a weighting sensitivity analysis of the composite $\mathscr{T}$ by varying $(w_{\text{ctx}}, w_{\text{gr}}, w_{\text{ans}})$. The average correlation with offline metrics was stable in the range 0.238–0.273. The answer-heavy setting $(0.25, 0.25, 0.5)$ achieved the highest alignment ($r_{\text{avg}} = 0.273$; +7.5% vs. baseline 0.254), while Equal weights yielded 0.261 (+2.8%). In contrast, Context-heavy dropped to 0.238 (–6.3%), and removing the answer term (Pure C+G) degraded alignment to 0.188 (–26.0%). These results confirm that the baseline (0.4/0.3/0.3) is balanced and robust, with answer relevance being the most critical component for mirroring offline quality.

S.V. Additional Smart Cabin Scenarios

We provide additional scenarios:

- **Emergency vehicle yielding (Michigan, USA)**, covering explicit right-of-way, lane positioning, and intersection-clearing requirements under state law (Figure S5)
- **Rural road interaction with horse-drawn carriages (Ontario, Canada)**, illustrating less common but legally relevant edge cases that frequently confuse non-local drivers (Figure S6)
- **Cross-border driving from Hong Kong to Mainland China**, involving changes in driving side, roundabout traversal rules, right-turn-on-red exceptions, expressway lane usage, and road-marking semantics (Figure S7)

References

[1] Z. Xiang, C. Wu, Q. Zhang, S. Chen, Z. Hong, X. Huang, J. Su, When to use graphs in rag: A comprehensive analysis for graph retrieval-augmented generation, arXiv preprint arXiv:2506.05690 (2025).

[2] Y. Zheng, R. Zhang, J. Zhang, Y. Ye, Z. Luo, Z. Feng, Y. Ma, Llamafactory: Unified efficient fine-tuning of 100+ language models, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations), Association for Computational Linguistics, Bangkok, Thailand, 2024. URL `http://arxiv.org/abs/2403.13372`

Country/region	Query	Retrieval GT	Answer GT
		Vector-retrieval	
ON, Canada	What concentration of alcohol in the blood can lead to a police officer requesting the surrender of a driver's license according to the Criminal Code (Canada)?	`file: en.pdf` excerpt: ... indicates that the concentration of alcohol in his or her blood is 50 milligrams or more of alcohol in 100 millilitres of blood, a police officer may request that the person surrender his or her driver's licence. ...	50 milligrams or more of alcohol in 100 millilitres of blood.
HK, China (Hong Kong SAR)	What is the maximum speed limit for medium and heavy goods vehicles?	`file: gt_hk.json` excerpt: ...Goods vehicles: the maximum speed limit for medium and heavy goods vehicles is 70 km/h...	70 km/h.
MI, USA	What is prohibited for individuals with a level 1 or level 2 graduated license while operating a motor vehicle on a highway or street?	`file: gt_mi.json` excerpt: ...shall not use a cellular telephone while operating a motor vehicle upon a highway or street; "use" means initiating/answering a call or engaging in verbal communication...	Using a cellular telephone to initiate a call, answer a call, or engage in verbal communication.

Table S4: Sample entries from the DriveLegal-RAG datasets (vector-retrieval).

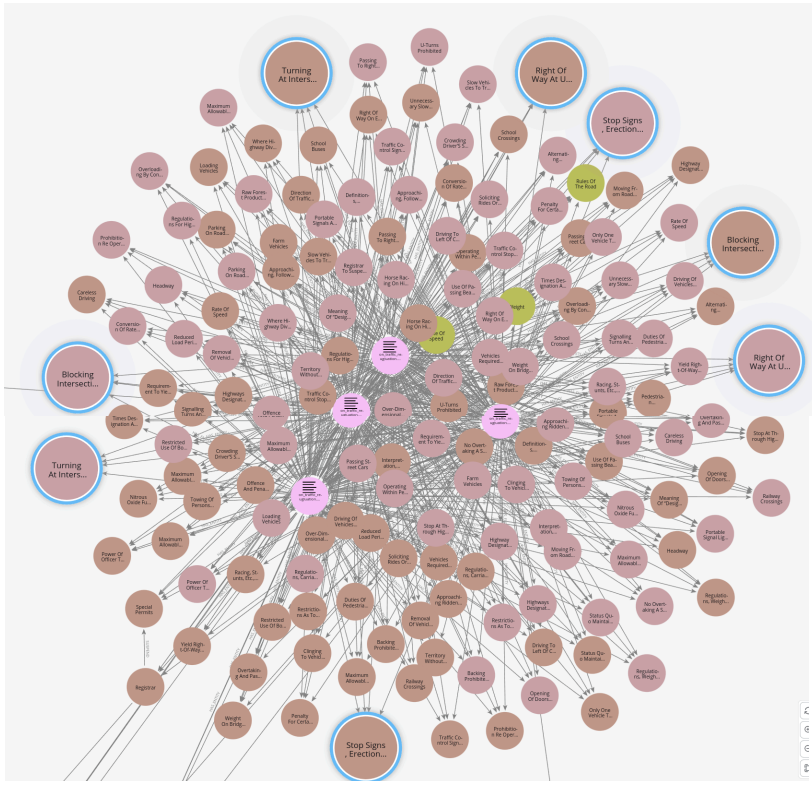

Figure S2: **Knowledge graph for Ontario, Canada driving regulations.**

Question Type	Query	Evidence	Evidence Relations	Answer GT
Fact Retrieval	What must a driver do if their permit is suspended?	the Registrar shall notify the permit holder of the proposed action	This quote indicates that the driver will be informed of any permit-related actions, guiding their response.	The driver must be aware that "the Registrar shall notify the permit holder of the proposed action."
Complex Reasoning	What should a driver do if a police officer signals for them to stop while another vehicle approaches from behind?	Every officer who detains a motor vehicle under this section shall, as soon as practicable.	The evidence emphasizes the obligation of the officer to act promptly when detaining a vehicle, indicating the driver's need to comply with the officer's signal immediately.	The driver should stop safely, yielding to the officer's signal. As stated, "Every officer who detains a motor vehicle under this section shall, as soon as practicable."
Contextual Summarize	What should drivers do when receiving a notice of suspension?	requiring the person to retain the records specified in the notice for the period of time specified in the notice. / A person to whom a notice is given shall comply with the provisions of the notice.	These quotes emphasize the driver's responsibility to keep necessary documents and follow the rules outlined in the notice for compliance.	Drivers must "retain the records specified in the notice" and "comply with the provisions of the notice" to ensure legal adherence.
Creative Generation	Imagine you are a driving instructor facing a scenario where a 15-year-old student wants to practice driving. Draft a detailed response addressing legal requirements for this situation.?	No person under the age of sixteen years shall drive or operate a motor vehicle... on a highway. / No person shall employ or permit anyone under the age of sixteen years to drive or operate a motor vehicle...	These legal requirements explicitly prohibit individuals under 16 from driving or being allowed to drive by instructors, emphasizing the importance of adhering to age restrictions for road safety.	In this scenario, as a driving instructor, it is imperative to adhere to the legal obligations surrounding underage driving. Firstly, according to R.S.O. 1990, c. H.8, s. 37 (1), "No person under the age of sixteen years shall drive or operate a motor vehicle... on a highway." Therefore, it is unlawful to allow the 15-year-old student to drive any motor vehicle, including a streetcar or tractor. Additionally, as the instructor, you must comply with R.S.O. 1990, c. H.8, s. 37 (2), which states, "No person shall employ or permit anyone under the age of sixteen years to drive or operate a motor vehicle..." Thus, you cannot allow or facilitate any driving practice for this student. Furthermore, regarding bicycles, R.S.O. 2009, c. 5, s. 8 prohibits allowing a person under 16 to operate a motor-assisted bicycle. To ensure compliance, you should structure your training program to focus on theoretical knowledge until the student reaches the legal driving age. This approach mitigates legal risks and promotes road safety.

Table S5: Sample entries from Ontario split of the DriveLegal-RAG datasets (graph-based RAG). Some entries are truncated.

Table S6: GraphRAG split word-length statistics (avg. words per field). We follow GraphRAG-Bench [1] format

Region	Entries	Question	Answer	Context (Evidence)
HK	500	16.79	25.17	37.81
MI	500	17.11	26.95	47.26
ON	500	16.84	26.67	48.00
UK	500	16.56	24.69	33.80
Total	**2000**	**16.83**	**25.87**	**41.72**

Table S7: Settings for Fine-tuning

Parameter	Value	Parameter	Value
Learning Rate	5e-5	Max New tokens	512
Cutoff Length	1024	Temperature	0.95
Top-p	0.7	Quantization	BitsandBytes

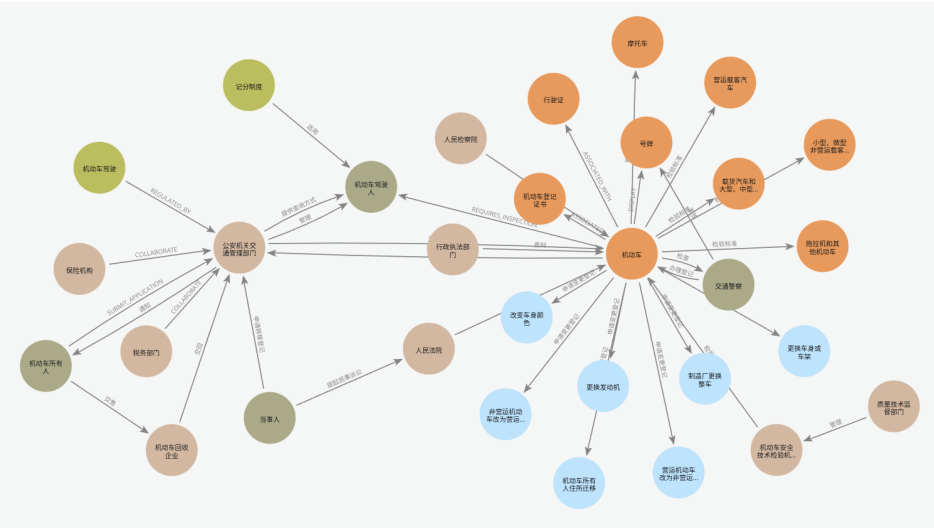

Figure S3: **Knowledge graph for China driving regulations.**

Table S8: Key LightRAG parameters used for all retrieval and reasoning experiments.

LightRAG Configuration Summary	
`Embedding Model`	`Qwen3-Embedding-0.6B`
`Query Type`	`naive / local / global / hybrid / mix`
`Construction LLM`	`GPT-4o-mini`
`Chunk Token Size`	1200
`Chunk Overlap`	100
`Retrieval Top-k`	5
`Max Tokens (Text Unit)`	4000
`Max Tokens (Global Context)`	4000
`Max Tokens (Local Context)`	4000
`Temperature`	0.7 (default)
`Cosine Threshold`	0.2 (semantic match filter)
`Max Graph Nodes`	1000

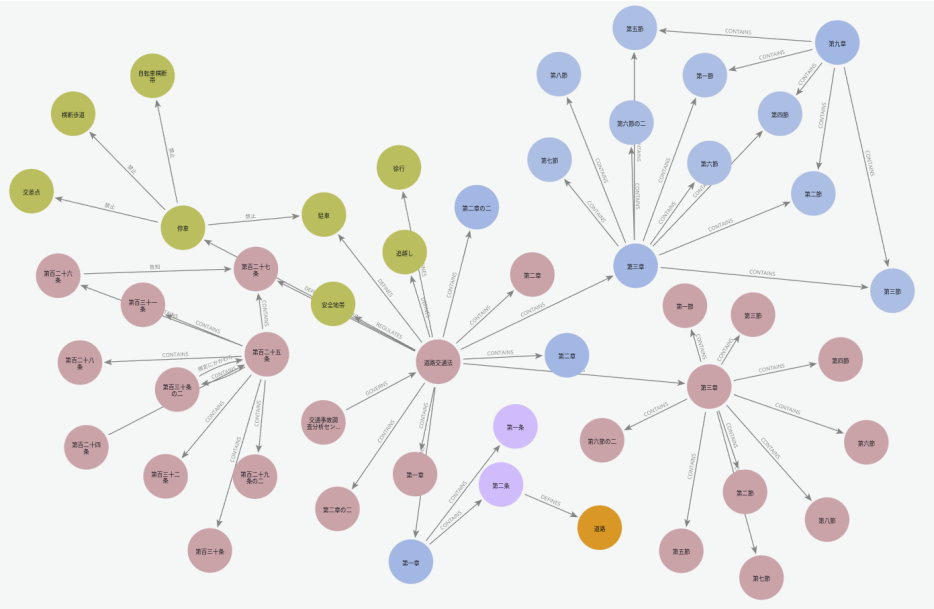

Figure S4: **Knowledge graph for Japan driving regulations.**

Table S9: Correlation with offline metrics under different triad weightings. Best average in **bold**; ablation in *italics*.

Weights (w_c, w_g, w_a)	Retr-Ctx	Retr-Rec	ROUGE	Ans-Corr	Coverage	Faithful	Avg
Baseline (0.40, 0.30, 0.30)	0.438	0.022	0.196	0.157	0.249	0.462	0.254
Equal (1/3, 1/3, 1/3)	0.429	0.029	0.201	0.172	0.252	0.481	0.261
Context-Heavy (0.50, 0.25, 0.25)	0.442	0.011	0.183	0.131	0.238	0.424	0.238
Grounded-Heavy (0.25, 0.50, 0.25)	0.379	0.036	0.218	0.161	0.235	0.459	0.248
Answer-Heavy (0.25, 0.25, 0.50)	0.426	0.040	0.188	0.207	0.260	**0.515**	**0.273**
Pure C+G (0.40, 0.30, 0.00)	0.370	-0.004	0.181	0.059	0.188	0.308	0.184

Scenario: Emergency Vehicle Yielding (Michigan, USA)
Input query: I'm visiting Michigan—if an ambulance or fire truck comes up behind me with lights/siren, what exactly am I supposed to do (pull over, stop, which side, and what if I'm at an intersection)?

Retrieved Context:
[Source: us.pdf, Page 61] *257.653 Immediate approach of authorized emergency vehicle; duty of driver of another vehicle.* Sec. 653. (1) Upon the immediate approach of an authorized emergency vehicle equipped with not less than 1 lighted flashing, rotating, or oscillating lamp exhibiting a red or blue light visible under normal atmospheric condition from a distance of 500 feet to the front of the vehicle and when the driver is giving audible signal by siren, exhaust whistle, or bell: (a) The driver of another vehicle shall yield the right of way and shall immediately drive to a position parallel to and as close as possible to the right-hand edge or curb of the roadway, clear of an intersection, and shall stop and remain in that position until the authorized emergency vehicle has passed, except when otherwise directed by a police officer... (3) A person who violates this section is responsible for a civil infraction. **[Source: us.pdf, Page 61]** *257.653a Stationary emergency vehicle giving visual signal.* Sec. 653a. (1) Upon approaching and passing a stationary authorized emergency vehicle that is giving a visual signal by means of flashing, rotating, or oscillating red, blue, white, or amber lights as permitted by section 698, the driver of an approaching vehicle shall exhibit due care and caution...

Response:
Pull over to the right-hand side of the road, stop, and remain stopped until the emergency vehicle has passed. If you're at an intersection, proceed through it safely and then pull over. Do not block the intersection.

Evaluation:

Metric	*Score*	*Reasoning*
Context Relevance	1.0	The retrieved context (Sec. 653) directly addresses the duty of a driver when an authorized emergency vehicle is approaching with lights and siren, including the requirement to pull to the right, stop, and clear intersections.
Groundedness	1.0	The response accurately reflects MCL Section 653 requirements: yielding right of way, moving to the right-hand edge, stopping clear of intersections, and remaining stopped until the vehicle passes. Penalty information is supported by Sec. 653(3) and 653a.
Answer Relevance	1.0	The response thoroughly addresses all aspects of the query: pulling over to the right, stopping, handling intersections, and consequences of non-compliance.

Figure S5: Emergency vehicle yielding scenario demonstrating retrieval of MCL 257.653 requirements. Retrieved contents and evaluation reasons are partially shown.

Scenario: Rural Road with Horse-Drawn Carriage (Ontario, Canada)
Input query: I'm driving in rural Ontario and I came up behind a horse-drawn carriage—what are the rules for passing it safely and legally, and what should I do if the horse looks spooked?

Retrieved Context :
[Source: ca.pdf, Page 233] *Approaching ridden or driven horses, etc. Section 167*: Every person having the control or charge of a motor vehicle or motor assisted bicycle on a highway, when approaching a horse or other animal that is drawing a vehicle or being driven, led or ridden, shall operate, manage and control the motor vehicle or motor assisted bicycle so as to exercise every reasonable precaution to prevent the frightening of the horse or other animal and to ensure the safety and protection of any person driving, leading or riding upon the horse or other animal or being in any vehicle drawn by the horse or other animal. R.S.O. 1990, c. H.8, s. 167. **[Source: ca.pdf, Page 223]** *Slow vehicles to travel on right side. Section 147(1)*: Any vehicle travelling upon a roadway at less than the normal speed of traffic at that time and place shall, where practicable, be driven in the right-hand lane then available for traffic or as close as practicable to the right hand curb or edge of the roadway. **[Source: ca.pdf, Page 223]** *Vehicles or equestrians overtaken. Section 148(2)*: Every person in charge of a vehicle or on horseback on a highway who is overtaken by a vehicle or equestrian travelling at a greater speed shall turn out to the right and allow the overtaking vehicle or equestrian to pass.

Response:
Slow down, maintain a safe distance, avoid sudden movements or loud noises, and do not pass if the horse appears spooked. Wait until the horse is calm and it is safe to proceed. The law requires you to exercise every reasonable precaution to prevent frightening the horse and ensure the safety of anyone riding or driving it.

Evaluation:

Metric	*Score*	*Reasoning*
Context Relevance	0.8	Section 167 directly addresses approaching horses with the duty to exercise reasonable precaution. Sections 147–148 provide supplementary overtaking rules. Minor gap: no explicit guidance on handling a spooked horse beyond the general duty.
Groundedness	1.0	The response accurately summarizes Section 167's requirements. Guidance on caution, safe distance, and waiting if unsafe are consistent with the statutory language requiring "every reasonable precaution."
Answer Relevance	1.0	The response addresses both parts of the query: safe/legal passing rules and specific guidance for spooked horses, with legally grounded recommendations.

Figure S6: Horse-drawn carriage scenario demonstrating retrieval of HTA Section 167 and related overtaking rules. Retrieved contents and evaluation reasons are partially shown.

Scenario: Cross-Border Driving (Hong Kong → Mainland China)

Input query: 我第一次从开车北上到内地（右车、内地靠右行驶），环岛要怎么走？红灯到底能不能右转？高速公路车道怎么选、限速怎么看？路面黄线/白线和'锯齿线'分别表示什么？

Translation: *This is my first time driving northbound into mainland China from Hong Kong (right-hand-drive vehicle, while the mainland drives on the right). How should I navigate roundabouts? Can I turn right at a red light or not? How do I choose lanes on expressways, and how do I read speed limits? What do yellow lines, white lines, and "zigzag lines" on the road mean?*

Retrieved Context:

[Source: 20240930_tc.pdf, Page 1–2] 与香港不同，内地的车是左车，靠右行驶。特别提醒：驾驶车辆进入环岛行驶时要注意行车方向：一切车辆只能靠右，逆时针行驶，且环内车有优先权。遇到红灯到底能不能右转？如遇到没有设置限制右转的信号灯，在礼让被放行的车辆、行人及非机动车的前提下，可右转弯。内地道路上黄色、白色标线：黄色的标线是分隔对向车辆；白色标线是分隔同向车辆。路面上的锯齿线：看到锯齿线，就不要开太快啦！这是提示你减速慢行。前方可能会有转弯、下坡或高速入口。高速公路上的车道从左到右依次为：超车道、行车道、慢车道和应急车道。高速公路一般最高车速不得超过每小时120公里，最低车速不得低于每小时60公里。

Translation: *Unlike Hong Kong, vehicles in mainland China are left-hand-drive and traffic keeps to the right. A special reminder: when driving into a roundabout, pay close attention to the direction of travel. All vehicles must keep to the right and travel counterclockwise, and vehicles already inside the roundabout have the right of way.*

So, can you turn right at a red light? If there is no signal or sign prohibiting right turns, you may turn right after yielding to vehicles, pedestrians, and non-motorized road users that have the right of way.

On mainland roads, yellow and white road markings have different meanings: yellow lines separate traffic flowing in opposite directions, while white lines separate traffic moving in the same direction. Zigzag lines painted on the road indicate that drivers should slow down. They warn of situations ahead such as curves, downhill sections, or expressway entrances.

On expressways, lanes from left to right are the passing lane, driving lane, slow lane, and emergency lane. The maximum speed on expressways is generally 120 km/h, and the minimum speed is usually 60 km/h.

[Source: northbound travel for hong kong vehicles.pdf, Page 1] 香港采用「左上右落」的道路设计，车辆靠左行驶；方向盘设于车辆右方。相反，内地采用「右上左落」的道路设计，车辆靠右行驶；方向盘设于车辆左方。当香港车辆在内地驾驶时，驾驶人士的视野范围会因道路系统的转变而受到局限及变得狭窄，盲区范围因此增大并带来较高意外风险。

Translation: *Hong Kong uses a "left-in, right-out" road design, with vehicles driving on the left and the steering wheel on the right. In contrast, mainland China uses a "right-in, left-out" road design, with vehicles driving on the right and the steering wheel on the left. When Hong Kong vehicles are driven in the mainland, drivers' fields of vision may be restricted due to the change in road system, increasing blind spots and thus raising the risk of accidents.*

[Source: hk.pdf, Page 59] Overtaking on dual carriageways: Overtake only on the right. Never move to a lane on the left to overtake. You must not use a hard shoulder for overtaking.

Response:

环岛必须左转进入，顺时针行驶，让右侧车辆先行。红灯时若无禁止标志可以右转，但须礼让行人。高速公路车道从左至右为：超车道、行车道、慢车道、应急车道；限速60–120 km/h。黄线分隔对向车流，白线分隔同向车流，锯齿线表示需减速慢行。

Translation: *At roundabouts, you must enter by turning left, drive clockwise, and yield to vehicles on your right. When the light is red, you may turn right if there is no sign prohibiting it, but you must yield to pedestrians. On expressways, the lanes from left to right are: passing lane, travel lane, slow lane, and emergency lane; the speed limit is 60–120 km/h. Yellow lines separate opposing traffic flows, white lines separate traffic moving in the same direction, and zigzag lines indicate you should slow down.*

Evaluation:

Metric	*Score*	*Reasoning*
Context Relevance	1.0	The "港车北上" (Northbound Travel for Hong Kong Vehicles) driving guide (20240930_tc.pdf) directly addresses all query aspects: roundabout rules, red-light right turns, highway lanes/speed limits, and road markings including sawtooth lines.
Groundedness	0.93	Most claims are directly supported: right-on-red conditions, highway lane order, speed limits, and line meanings are accurately reflected. Minor discrepancy in roundabout phrasing (clockwise vs. counterclockwise terminology).
Answer Relevance	1.0	The response comprehensively addresses all five sub-questions: roundabout navigation, right-on-red rules, lane selection, speed limits, and road marking interpretations for cross-border drivers.

Figure S7: Cross-border driving scenario demonstrating multi-document retrieval for jurisdictional transition (HK→Mainland China). Retrieved contents and evaluation reasons are partially shown.